

DPP NO.-7

- Q.1 There are 10 red balls of different shades & 9 green balls of identical shades. Then the number of arranging them in a row so that no two green balls are together is
(A) $(10!) \cdot {}^{11}P_9$ (B) $(10!) \cdot {}^{11}C_9$ (C) $10!$ (D) $10! \cdot 9!$
- Q.2 Number of ways in which n distinct objects can be kept into two identical boxes so that no box remains empty, is _____.
- Q.3 A shelf contains 20 different books of which 4 are in single volume and the others form sets of 8, 5 and 3 volumes respectively. Number of ways in which the books may be arranged on the shelf, if the volumes of each set are together and in their due order is
(A) $\frac{20!}{8! \cdot 5! \cdot 3!}$ (B) $7!$ (C) $8!$ (D) $7 \cdot 8!$
- Q.4 If all the letters of the word "QUEUE" are arranged in all possible manner as they are in a dictionary, then the rank of the word QUEUE is :
(A) 15^{th} (B) 16^{th} (C) 17^{th} (D) 18^{th}
- Q.5 Number of rectangles in the grid shown which are not squares is
(A) 160 (B) 162
(C) 170 (D) 185
- | | | | | | |
|--|--|--|--|--|--|
| | | | | | |
| | | | | | |
| | | | | | |
| | | | | | |
- Q.6 All the five digit numbers in which each successive digit exceeds its predecessor are arranged in the increasing order of their magnitude. The 97^{th} number in the list does not contain the digit
(A) 4 (B) 5 (C) 7 (D) 8
- Q.7 The number of combination of 16 things, 8 of which are alike and the rest different, taken 8 at a time is _____.
- Q.8 The number of different ways in which five 'dashes' and eight 'dots' can be arranged, using only seven of these 13 'dashes' & 'dots' is :
(A) 1287 (B) 119 (C) 120 (D) 1235520
- Q.9 In a certain college at the B.Sc. examination, 3 candidates obtained first class honours in each of the following subjects: Physics, Chemistry and Maths, no candidates obtaining honours in more than one subject; Number of ways in which 9 scholarships of different value be awarded to the 9 candidates if due regard is to be paid only to the places obtained by candidates in any one subject is _____.
- Q.10 There are n identical red balls & m identical green balls. The number of different linear arrangements consisting of " n red balls but not necessarily all the green balls" is ${}^x C_y$ then
(A) $x = m + n, y = m$ (B) $x = m + n + 1, y = m$ (C) $x = m + n + 1, y = m + 1$ (D) $x = m + n, y = n$

Direction for Q.11 & Q.12

In how many ways the letters of the word "COMBINATORICS" can be arranged if

- Q.11 All the vowels are always grouped together to form a contiguous block.
- Q.12 All vowels and all consonants are alphabetically ordered.
- Q.13 How many different arrangements are possible with the factor of the term $a^2b^4c^5$ written at full length.
- Q.14 Find the number of 4 digit numbers starting with 1 and having exactly two identical digits.
- Q.15 Number of ways in which 5 A's and 6 B's can be arranged in a row which reads the same backwards and forwards, is